

Master Data Asset Index

Overview

This document serves as a comprehensive index of all data assets held by our advisory practice. Each asset is classified by type, source methodology, temporal coverage, and strategic value.

Category 1: Field Deployment Data (India)

Asset	Description	Coverage	Records
Paired Crop-Cut Yield Data	Per-farmer baseline vs. treatment yield observations	216 farmers, 42 villages, 5 zones	216 paired observations
Site Deployment Parameters	Application rates, total tonnage, area covered per zone	6,937 hectares	5 deployment zones
Economic Impact Analysis	MSP-adjusted income calculations across 3 Indian states	Kharif 2025 season	Per-farmer line items
Geological Source Data	Basalt provenance (Deccan Traps, Rajmahal Traps)	2 geological formations	Feedstock chemistry available
Yield-Productivity Correlation	Statistical analysis of yield response vs. baseline soil quality	$r = -0.41$	Quartile-level breakdown

Key Metrics:

- Total basalt tracked: 145,157 tonnes
- Median yield increase: +24.7%
- Q1 (poorest soil) yield gain: +47%
- Farmer income uplift: 207–443/season

Category 2: Controlled Lysimeter Data (Europe)

Asset	Description	Coverage	Records
Leachate Chemistry (Major Ions)	Ca ²⁺ , Mg ²⁺ concentrations per sampling interval	777+ days (2022–2024)	~100 time-series records per treatment
Electrical Conductivity Logs	In-situ EC sensor data at multiple soil depths	Continuous	Daily resolution
Leachate Volume Data	Volume of water exported per sampling interval	Per sampling event	All 5 treatments
CO ₂ Concentration Data	Soil, tank, and ambient air CO ₂ levels	2022–2024	Multi-depth
Irrigation & Weather Data	Rainfall (mm), irrigation timing, water balance	Full experiment duration	Per-event logging

Key Metrics:

- 5 treatment arms: Control, 100 t/ha, 200 t/ha, 400 t/ha, FINE
- ⁴⁰⁰/₁₀₀ scaling ratio: −58.6× (expected: 4.0×)
- FINE treatment excess Ca: +85,200 μmol (vs. −333,000 for standard 200 t/ha)

Category 3: Soil & Feedstock Characterization

Asset	Description	Parameters
XRF Elemental Analysis	Full oxide composition of deployment soils	SiO ₂ , Al ₂ O ₃ , Fe ₂ O ₃ , MgO, CaO, K ₂ O, etc.
Cation Exchange Capacity	Soil CEC measurements	meq/100g
TOC/TIC	Total organic and inorganic carbon baselines	%
Particle Size Distribution	Full laser diffraction PSD (0.55–763 µm)	Cumulative % passing
Trace Metals	Heavy metal screening (Cr, Co, Cu, Ni, V)	ppm
Feedstock PSD	Basalt grain size distribution	d10, d50, d90, mean, mode
BET Surface Area	Specific surface area of feedstock	m ² /g

Category 4: Derived Analytical Products

Asset	Description	Methodology
Passivation Point Identification	Exact temporal point where dissolution permanently stalls	Time-series inflection analysis
Non-Linearity Proof	Statistical proof that dose-response breaks at high application rates	Ratio analysis + delta plotting
Baseline Reversion Charts	Visual proof that Year 2+ reverts to control levels	Delta Ca export vs. time
Crusting Risk Model	Prediction of pore clogging based on grain size and dose	Hydraulic conductivity correlation
Year 1 Flush Characterization	Temporal profile of the initial weathering burst	Ion flux time-series
Poor Soil Premium Model	Yield prediction based on baseline soil degradation quartile	Inverse correlation model
Break-Even Calculator	Logistics cost threshold for profitability	Integrated carbon-economics model

Category 5: Academic & Regulatory Intelligence

Asset	Description	Sources
Passivation Layer Literature	Peer-reviewed evidence for silica/iron dissolution barriers	ACS, ScienceDirect, AGU
Hydraulic Conductivity Synthesis	Global meta-analysis of ERW impacts on soil hydrology	Independent academic synthesis
Carbon Registry PDD Templates	Project Design Document frameworks for Puro.earth/Isometric	Registry guidelines
Competitor MRV Analysis	Reverse-engineered MRV methodologies from public registry filings	Public PDDs

Data Security Classification

Tier	Contents	Sharing Policy
Tier 1 — Public Teaser	Surface-level results, aggregate statistics, methodology overview	Shared freely in pitch materials
Tier 2 — NDA Required	Site-level data, soil characterization, feedstock specs	Shared under signed NDA
Tier 3 — JV Partners Only	Kinetic models, geographic targeting maps, calibration coefficients	Shared only with contracted JV partners
Tier 4 — Internal Only	Raw datasets, source partner identities, proprietary algorithms	Never shared externally